



AI & Software Engineering Workshop

Week 2 — Day 1: Setup and Overview

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What you'll build this week

- A 350-million-parameter Armenian language model
- Trained **from scratch** — no pretrained weights, no shortcuts
- Fine-tuned into a **chatbot**
- Deployed on **HuggingFace**
- Plugged into **your Week 1 Telegram bot**

Last week you rented a brain. This week you grow one.

What *is* a language model?

- A function that answers one question: "**what token comes next?**"
- Trained by reading text and predicting the next word, billions of times
- Wrong guess → adjust the weights a tiny bit → repeat
- Everything else — chat, translation, jokes — falls out of that one skill

GPT, Claude, Llama: same core idea, more zeros in the parameter count.

The pipeline — and the repo

1_download.py	get the corpus (or pre-tokenized data)	
2_prepare.py	clean + dedup	(done for you)
3_tokenize.py	text → numbers (BPE)	
4_train.py	pretrain	(Days 3–4)
5_generate.py	sample text from a checkpoint	
6_finetune.py	teach it to chat	(Day 5)
7_deploy.py	upload to HuggingFace	(Day 5)
8_chat.py	talk to your model	

github.com/EdikSimonian/armenian-gpt

Setup

```
git clone https://github.com/EdikSimonian/armenian-gpt.git
cd armenian-gpt
pip install numpy sentencepiece huggingface_hub zstandard
pip install torch --index-url https://download.pytorch.org/whl/cu121
```

HuggingFace: create an account, then a **read token**:

```
export HF_TOKEN=hf_...
```

Demo: train a model in 2 minutes

```
python 4_train.py --preset tiny --tokenizer bpe  
python 5_generate.py --prompt "Հայաստանը"
```

- ~0.2M parameters, 1000 steps, under 2 minutes
- The output is... not great. **That's the point.**

Tiny vs. the real thing

Open the live demo: huggingface.co/spaces/edisimon/armgpt-demo

Same code. Same data. Same training loop.

	tiny (just now)	demo model
Parameters	~0.2M	~1B
Training	2 minutes	days
Output	word soup	real Armenian

The *only* differences: *size* and *time*. That's the lesson of the week.

The menu

Preset	Params	Steps	~Time (RTX 4090)
tiny	0.2M	1K	< 1 min
small	10M	5K	~10 min
medium	30M	10K	~1 hr
large	85M	20K	~3 hrs
xlarge	350M	36K	~8 hrs ← ours
giant	~1B	122K	~48 hrs (not this week)

Today → Tomorrow

Today: the pipeline end-to-end, and a tiny model that proves it runs.

Tomorrow: the fuel. 63 GB of Armenian text — where it came from, and how text becomes numbers.